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Calculation (spiral points)

Calculation (spiral points)

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On the previous page you learned about saddle points, stable nodes and unstable nodes. On this page you will encounter even more types of equilibrium points.

Exercise A

1.0/1.0 point (graded)

Our system of differential equations for the rainbowfish and gouramis has also an equilibrium point at **(31.25, 12.03)**. Calculate the eigenvalues of the Jacobian matrix

$J \left(\begin{bmatrix} 31.25 \\ 12.03 \end{bmatrix} \right)$ and classify the equilibrium point.

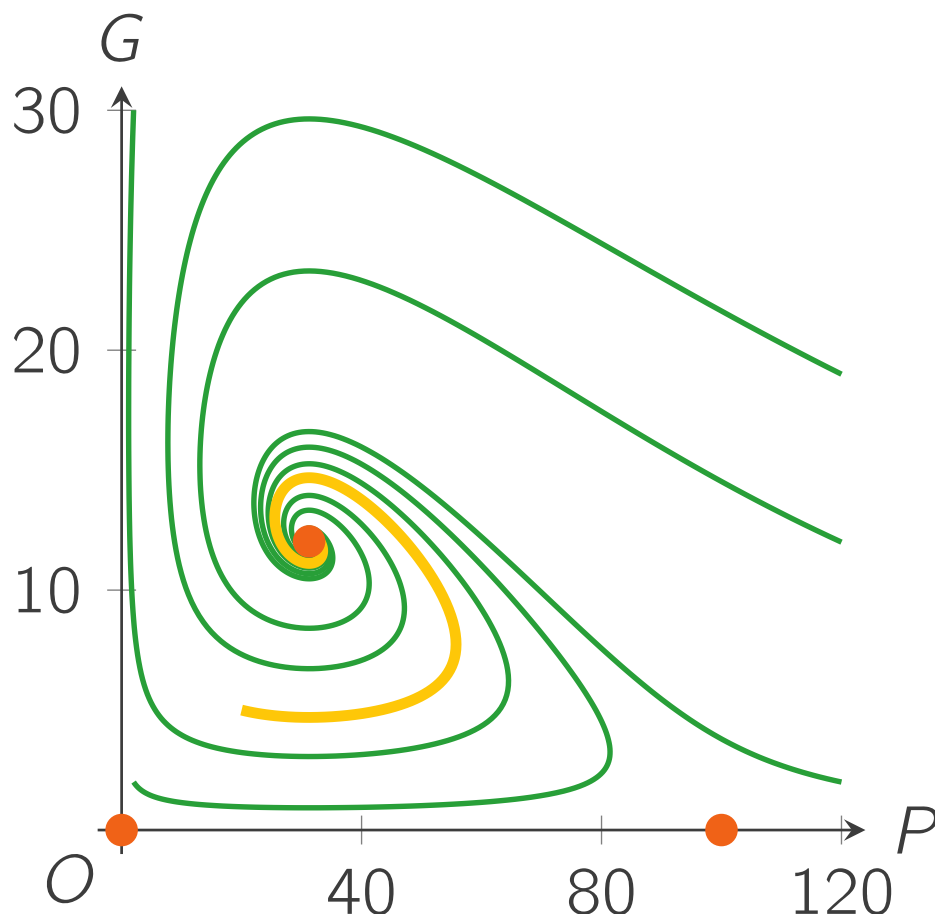
The eigenvalues indicate that equilibrium point **(31.25, 12.03)** is

- ☐ a stable node;
- ☐ a saddle point;
- ☐ an unstable node;
- ☒ something else. ✓

Submit

You have used 1 of 3 attempts

As you have seen in the last exercise, the eigenvalues of the Jacobian matrix in the point **(31.25, 12.03)** are complex. Complex numbers are not positive or negative, so this equilibrium point is neither a stable node, nor an unstable node nor a saddle point. So what is it then?



In the phase portrait, you see that the solutions near the equilibrium **(31.25, 12.03)** spiral around it. Can we explain this with the complex eigenvalues you found?

Consider the general form of the solution of $\vec{M}$ from the video:

$$\vec{M}(t) = c_1 e^{\lambda_1 t} \vec{v}_1 + c_2 e^{\lambda_2 t} \vec{v}_2.$$

The λ 's are complex numbers and the solution contains factors $e^{\lambda t}$. For exponential functions of complex numbers, we can use Euler's formula

$$e^{i\phi} = \cos(\phi) + i \sin(\phi).$$

Take the first eigenvalue λ_1 , which is complex of the form $\lambda_1 = a + bi$. Then

$$e^{\lambda_1 t} = e^{at+ibt} = e^{at} (\cos(bt) + i \sin(bt)).$$

($\lambda_2 = \overline{\lambda_1} = a - bi$ gives a similar expression.) The solution $\vec{M}(t)$ is now an expression with the complex i in it many times. Of course, $\vec{M}$ denotes a vector with (shifted) population sizes, so it should be real. In this course, we will not go into the details of how you can rewrite the solution for $\vec{M}$ to a real expression, but just state the result:

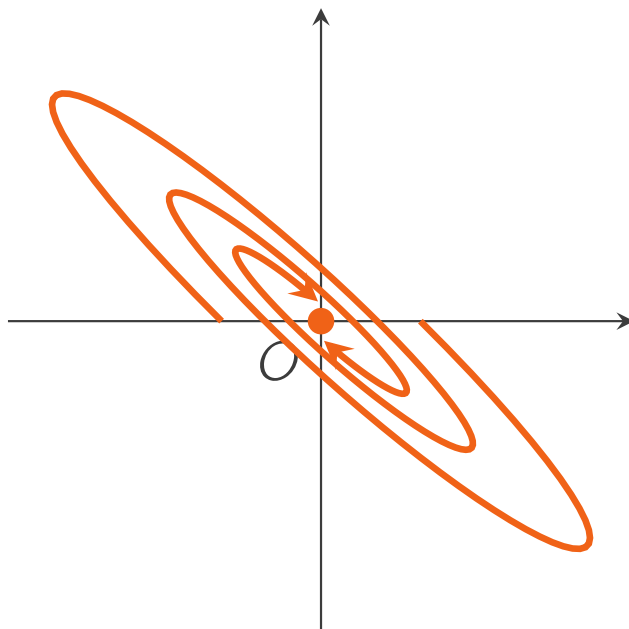
When the eigenvalues of the Jacobian matrix are complex numbers: $\lambda = a \pm bi$, the solutions for $\vec{M}(t)$ contain factors $e^{at} \cos(bt)$ and $e^{at} \sin(bt)$.

Looking at these factors, we see that each component of $\vec{M}(t)$ has oscillating parts: either $\cos(bt)$, or $\sin(bt)$ or both. So all terms oscillate with an angular frequency of b radians per day, where b is the imaginary part of the eigenvalues. The graphs of the solutions versus time oscillate up and down. In the phase portrait the trajectories will circle the origin.

Each component of $\vec{M}(t)$ also contains a factor e^{at} . If $a < 0$ that factor decreases in time, so the oscillations will be damped and in phase space the trajectories will spiral in towards the origin. If $a > 0$, the exponential factor will increase in time, making the oscillations larger and larger, and in the phase plane the trajectories will spiral outwards.

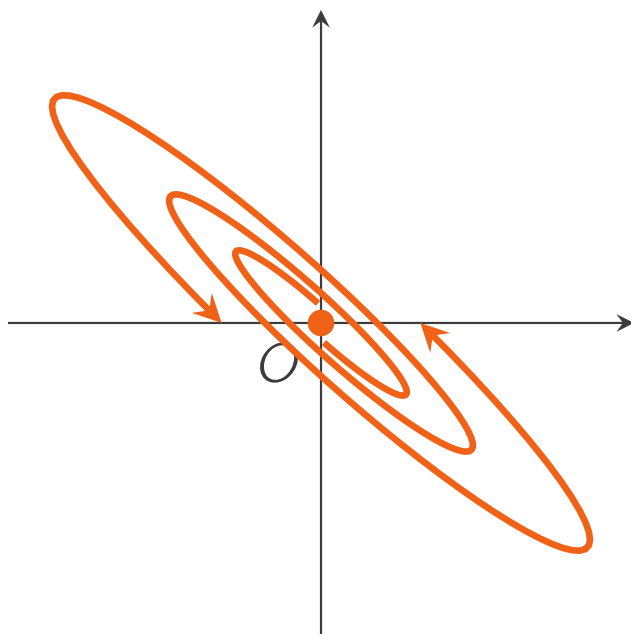
This type of behavior around an equilibrium point makes it a spiral point. When the real part a of the eigenvalue is negative, so $a < 0$, all solutions that start near the origin will end up in the origin, so that spiral point is a stable equilibrium point. When $a > 0$, solutions that start near the origin will spiral away from the origin, so then the origin is called an unstable spiral point.

An equilibrium point $\vec{X}_0$ is called a *stable spiral point* if the Jacobian matrix $J(\vec{X}_0)$ has two complex eigenvalues $\lambda = a \pm bi$ with negative real parts: $a < 0$. A typical sketch of the solutions near a stable spiral point in the phase plane is given by



$$\lambda = a \pm bi, a < 0$$

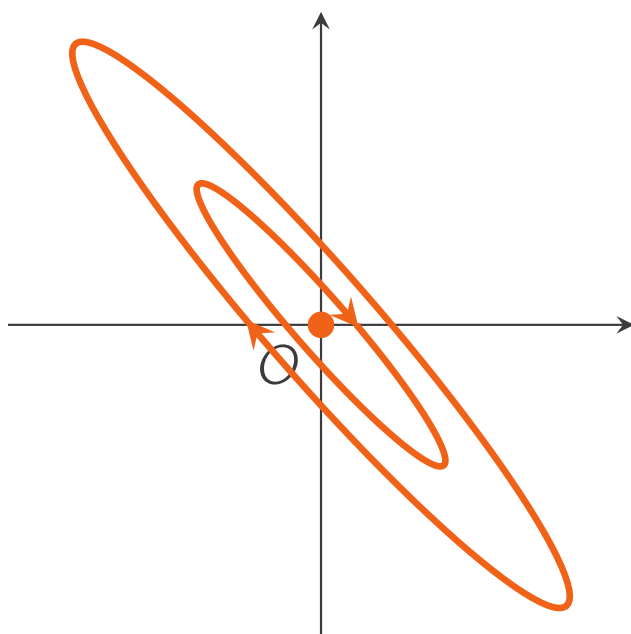
An equilibrium point $\vec{X}_0$ is called an *unstable spiral point* if the Jacobian matrix $J(\vec{X}_0)$ has two complex eigenvalues $\lambda = a \pm bi$ with positive real parts: $a > 0$. A phase portrait of an unstable spiral point is given in the following graph.



$$\lambda = a \pm bi, a > 0$$

When $a = 0$, the solution does not spiral inwards nor outwards, so the solution will be periodic, and in phase space the trajectory is a closed curve. Even though that curve is most often an ellipse, the equilibrium point is called a *circle point*.

An equilibrium point $\vec{X}_0$ is called a *circle point* if the Jacobian matrix $J(\vec{X}_0)$ has two complex eigenvalues with zero real parts: $\lambda = \pm bi$.



$$\lambda = \pm bi$$

Besides the six types of stability for equilibrium points discussed here, other types of stability exist, but these transcend the scope of this course.

Exercise E (1 point possible)

Question

The third equilibrium point (31.25,12.03) is



Answer



Reflection



Results

Correct Answer

a stable spiral point, because...

The real part of the eigenvalues is -0.11 and the imaginary part is plus or minus 0.33. This indicates it is a stable spiral point because of the negative real part and the nonzero imaginary part.

Your initial answer: a stable spiral point, because... (Option 1)

👤 "The Jacobian matrix has two complex eigenvalues with negative real parts (for both the eigenvalues the real part is -0.10935)"

Your final answer: a stable spiral point, because... (Option 1)

👤 "The Jacobian matrix has two complex eigenvalues with negative real parts (for both the eigenvalues the real part is -0.10935)"

See how your classmates answered below

Class Breakdown

This is a look at how your classmates answered the question during the initial and final rounds.

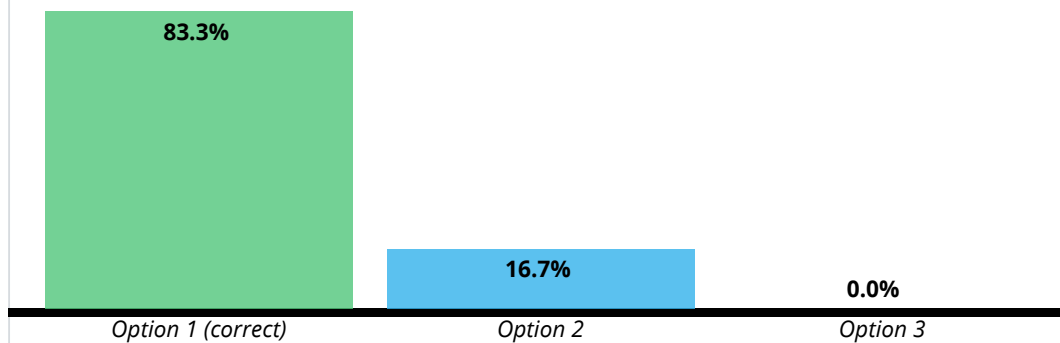
Answer Options

Option 1 (correct)
a stable spiral point, because...

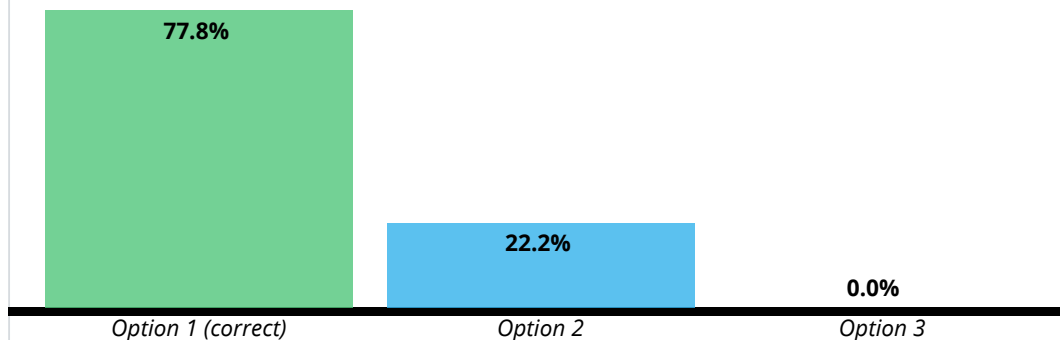
Option 2
an unstable spiral point, because...

Option 3
a circle point, because...

Initial Answer Selection



Final Answer Selection



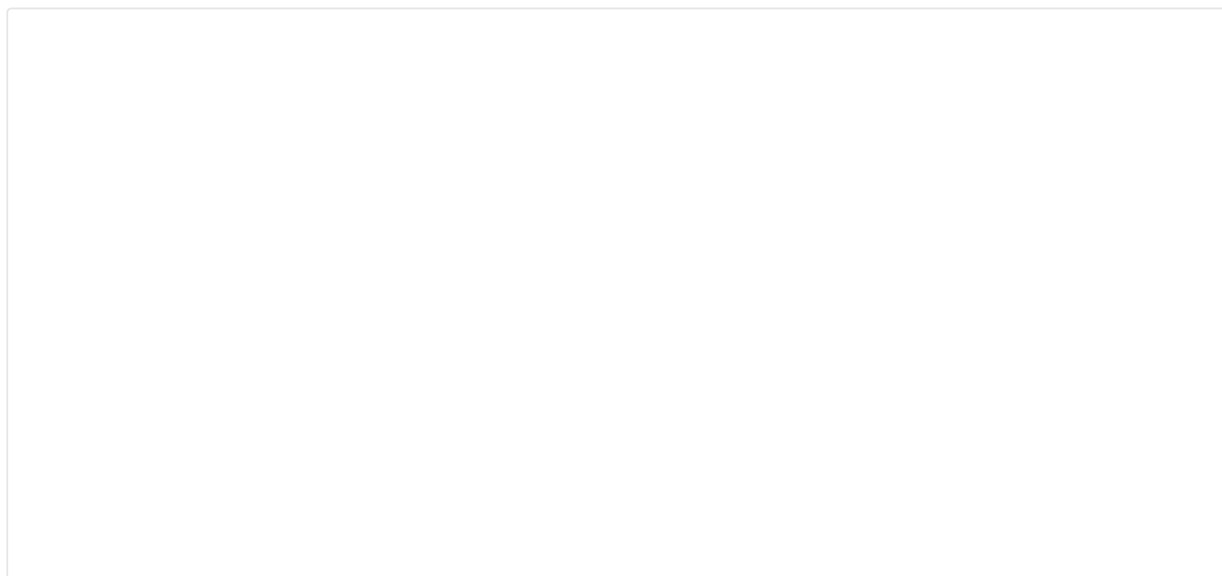
Exercise F

1/1 point (graded)

 Keyboard Help

PROBLEM

Previously we gave sketches in the phase plane of each of the six stability types. You can also make sketches of the solutions by drawing the graph of the components of the solution vector versus time. Match the following of these sketches to the given sketches in the phase plane.



Phase plane	Components vs. time	Phase plane	Components vs. time

Submit

You have used 1 of 3 attempts.

Reset

Show Answer

FEEDBACK

✓ Correctly placed 6 items.

✓ Good work! You have identified the component versus time graphs with the correct phase planes.

✓ Your highest score is 1.0

Were you surprised to see complex numbers turning up all of a sudden here? Did you have to brush up those skills from long ago? We think that complex numbers are taught in Calculus courses just for differential equations (and the exponential function as well), but we might be prejudiced. ;-)

OK, next we will validate the results from Euler's method with the conclusions about the equilibrium points.

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🌐 English ▼

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